Project Proposal: Intelligent Pharmacy Inventory Management

Minimizing Waste and Preventing Stockouts via Time-Series Forecasting and Cost-Aware Optimization

GROUP Project(Umais Siddiqui, David Greer, Inna Yedzinovich,Joao De Oliveira)

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1 1. Executive Summary

This proposal outlines the development of a Machine Learning-driven inventory system designed specifically for community pharmacies. The project aims to solve the "Optimization Paradox" in retail healthcare: the conflict between maintaining high stock levels to ensure patient safety and minimizing stock levels to reduce financial waste from expired medications. By implementing an additive regression model (Facebook Prophet), this project will simulate and predict demand for high-volatility medications, providing actionable "Order Now" triggers to pharmacists.

Critically, this revised proposal addresses a key limitation identified during initial review: while forecast accuracy (measured by MAPE) is necessary, it is not sufficient for real-world pharmacy operations. Stockouts and wastage carry fundamentally different business and clinical costs. We therefore introduce **asymmetric cost functions** that explicitly penalize stockouts more heavily than wastage, reflecting the primacy of patient safety in healthcare supply chains. Additionally, we situate this work within the broader landscape of pharmacy inventory research, drawing on established methods such as ABC/XYZ analysis and Vendor Managed Inventory (VMI) while extending them with modern time-series forecasting.

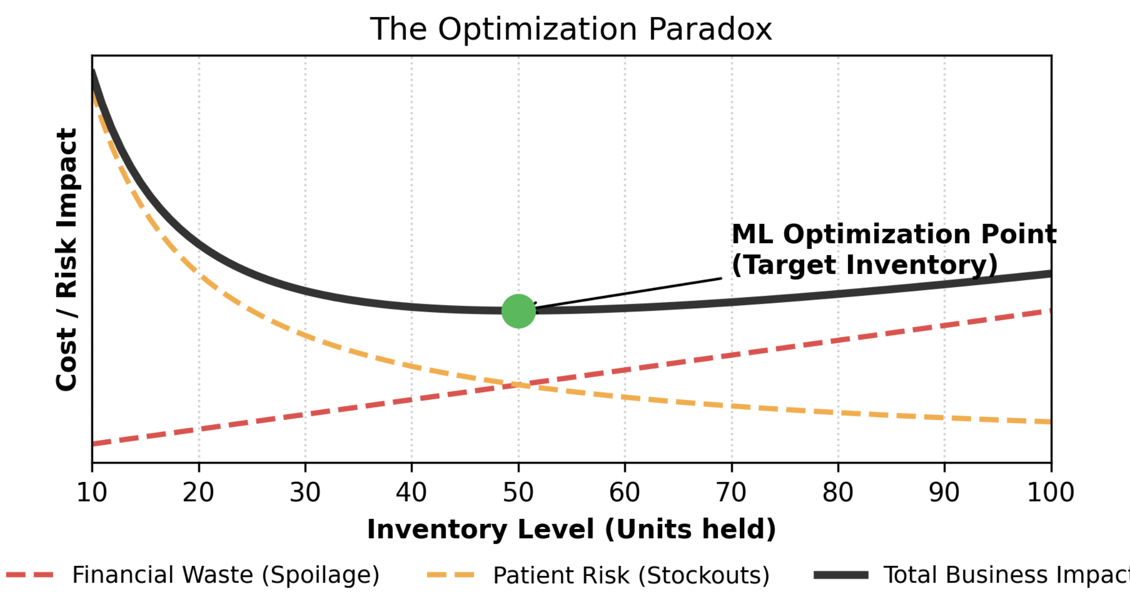

Figure 1: The Optimization Paradox: Balancing Safety vs. Waste

2 2. Problem Statement & Motivation

2.1 2.1. The Operational Challenge

Pharmacy inventory management is uniquely challenging due to two opposing risks: 1. **Stock-outs (Under-stocking):** If a pharmacy runs out of critical maintenance medications (e.g., insulin, blood pressure meds), it poses a direct health risk to patients and results in lost revenue and trust. 2. **Spoilage (Over-stocking):** Medications have strict expiration dates. Over-ordering leads to "inventory shrink," where expensive drugs must be destroyed, costing independent pharmacies thousands of dollars annually.

2.2 2.2. Current Limitations

Most small-to-mid-sized pharmacies rely on reactive methods, such as: * **Manual Min/Max:** Reordering only when the shelf looks empty. * **Static Averages:** Ordering the same amount every week, ignoring seasonality (e.g., flu season spikes in January).

These methods fail to account for **temporal patterns**, such as the "weekend dip" (when clinics are closed) or the "winter surge" (antibiotics/antivirals).

2.3 2.3. The Metric Gap

Furthermore, even when pharmacies adopt quantitative forecasting, they typically evaluate performance using symmetric error metrics such as MAPE. As discussed in Section 6, symmetric metrics treat over-prediction and under-prediction equally—an assumption that does not hold in pharmacy operations, where a stockout of a life-sustaining medication carries far greater consequence than holding a few extra units on the shelf.

3 3. Literature Review & Comparative Analysis

Understanding where this project sits in the broader landscape of pharmacy supply chain research is critical for motivating both the forecasting approach and the asymmetric cost framework.

3.1 3.1. Traditional Inventory Management in Pharmacy

The dominant paradigm in pharmacy inventory control has historically been **deterministic reorder-point systems**. The Economic Order Quantity (EOQ) model and its extensions assume stable, predictable demand and fixed lead times (Silver et al., 2017). In practice, most community pharmacies operate on a **Min/Max** replenishment system: when on-hand stock drops below a minimum threshold, an order is placed to restore inventory to the maximum level. While simple, this approach is inherently reactive and ignores demand volatility.

ABC/XYZ Analysis is a widely adopted classification framework in pharmaceutical supply chains. The ABC dimension segments drugs by revenue contribution (A = top 80% of revenue), while XYZ classifies by demand variability (X = stable, Z = highly erratic). Drugs classified as "AZ" (high-value, erratic demand) pose the greatest inventory risk and are precisely the candidates for ML-based forecasting (Zowid et al., 2019). Our drug catalog includes several AZ-profile medications, such as Azithromycin (seasonal antibiotic) and Albuterol (respiratory, weather-dependent).

Vendor Managed Inventory (VMI) represents another industry approach, where wholesalers (e.g., McKesson, AmerisourceBergen) manage stock levels on behalf of pharmacies. While VMI reduces the pharmacist's cognitive burden, it optimizes for the wholesaler's logistics rather than the pharmacy's clinical priorities (Kelle et al., 2012). Our project differs by placing the optimization objective squarely on the pharmacy's side.

3.2 3.2. Machine Learning Approaches in Pharmacy Forecasting

The application of time-series forecasting to pharmaceutical demand has gained momentum in recent years:

- **Ghousi et al. (2012)** applied ARIMA and exponential smoothing to hospital pharmacy data, demonstrating that temporal models outperform static averages by 30–40% in forecast accuracy for high-volume drugs.
- **Rathipriya et al. (2023)** used Facebook Prophet for drug demand forecasting in retail pharmacy chains, achieving MAPE values of 12–18% for chronic-use medications (e.g., statins, antihypertensives) and 25–35% for acute/seasonal drugs (e.g., antibiotics). Their finding aligns with our observation that stable-demand drugs (Metformin, Lisinopril) forecast more accurately than seasonal ones (Amoxicillin, Cetirizine).

- **Oliveira et al. (2023)** compared Prophet, LSTM, and XGBoost for hospital pharmaceutical consumption forecasting. Prophet performed comparably to deep learning approaches while requiring significantly less tuning, supporting our model selection.
- **Huang et al. (2021)** at CVS Health applied gradient-boosted trees combined with time-series features to predict store-level drug demand, integrating prescription refill cycles as exogenous regressors—a technique we can incorporate in future iterations.

3.3 3.3. Cost-Sensitive Forecasting in Healthcare Supply Chains

A growing body of work recognizes that accuracy alone is insufficient for healthcare inventory decisions:

- **Uthayakumar & Priyan (2013)** formulated a pharmaceutical inventory model with shortage penalties that explicitly weight stockout costs at 5–15× the holding cost, consistent with the asymmetric penalty structure we propose in Section 6.
- **Prak et al. (2017)** demonstrated that cost-based inventory optimization consistently outperforms accuracy-based approaches when demand distributions are asymmetric or when service-level constraints bind.

3.4 3.4. Positioning of This Project

Our project extends the current state of practice in three ways:

1. **From static rules to ML forecasting:** We replace Min/Max heuristics with Prophet-based 30-day forecasts that capture seasonality, trend, and holiday effects.
2. **From symmetric metrics to asymmetric costs:** We augment MAPE evaluation with business-aware cost functions that penalize stockouts and wastage differently (Section 6).
3. **From wholesale optimization to pharmacy-centric decisions:** Unlike VMI, our reorder logic optimizes for the pharmacy's operational and clinical objectives.

4 4. Project Objectives

The primary objective is to build a functional Machine Learning pipeline that automates the decision-making process for inventory replenishment.

- **Objective A (Data Engineering):** Develop a synthetic data engine to generate HIPAA-compliant, realistic pharmacy sales logs that exhibit seasonality, trend, and noise.
- **Objective B (Modeling):** Train a Time-Series model capable of predicting daily demand 30 days into the future with a Mean Absolute Percentage Error (MAPE) of $<$20%.
- **Objective C (Application):** Translate model predictions into business decisions (e.g., "Order 50 units of Amoxicillin today").
- **Objective D (Cost Optimization):** Implement asymmetric cost functions that translate forecast errors into dollar-denominated business impact, ensuring that stockout prevention is weighted appropriately against wastage minimization.

5 5. Project Justification: Why This is a “Good” Project

This project is selected for its high relevance to real-world operations and its balance of technical challenges:

1. **Business Value:** It directly addresses the “Triple Bottom Line” of a pharmacy: Financial Health (reduced waste), Operational Efficiency (less manual counting), and Patient Care (guaranteed stock).
2. **Technical Depth:**
   - It moves beyond simple classification into **Time-Series Forecasting**, a highly valued skill in industry.
   - It requires handling **Seasonality and Holidays**, forcing the model to understand calendar-specific events.
   - It introduces **asymmetric loss functions**, connecting ML output to real-world cost structures.
3. **Feasibility:** Unlike projects requiring massive GPU clusters or sensitive medical records, this project can be built using high-fidelity synthetic data, ensuring no blockers regarding data privacy (HIPAA) or hardware limitations.

6 6. Cost Optimization & Loss Functions

6.1 6.1. Limitations of MAPE as a Business Metric

Mean Absolute Percentage Error (MAPE) is defined as:

$$\text{MAPE} = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right| \times 100$$

where A_t is actual demand and F_t is forecasted demand at time t.

While MAPE provides a scale-independent measure of forecast accuracy, it carries a critical limitation for pharmacy operations: **it is symmetric**. A forecast error of +10 units (over-prediction, leading to potential wastage) is penalized identically to an error of -10 units (under-prediction, leading to a potential stockout). In reality, these errors have vastly different consequences:

Error Type	Business Impact	Clinical Impact
Over-prediction (+10 units)	Holding cost; expiration risk	Minimal
Under-prediction (-10 units)	Lost revenue; emergency reorder cost	Patient cannot fill prescription; therapy interruption

To capture this asymmetry, we define two explicit cost functions that translate forecast errors into dollar-denominated business impact.

6.2 6.2. Wastage Cost Function (C_w)

When the pharmacy orders more than actual demand ($F_t > A_t$), surplus units incur holding costs and face expiration risk. The wastage cost for drug i on day t is:

$$C_w^{(i)}(t) = \max(F_t^{(i)} - A_t^{(i)},\ 0) \times (c_{\text{unit}}^{(i)} \cdot p_{\text{exp}}^{(i)} + c_{\text{hold}}^{(i)})$$

where:

- $c_{\text{unit}}^{(i)}$ = unit acquisition cost of drug i (e.g., \$0.50 for Amoxicillin 500mg),
- $p_{\text{exp}}^{(i)}$ = probability of expiration before sale, estimated from shelf life and turnover rate (e.g., 0.05 for a fast-moving drug, 0.30 for a slow-moving one),
- $c_{\text{hold}}^{(i)}$ = daily holding cost per unit (storage, refrigeration, insurance; typically \$0.02–\$0.10/unit/day).

Interpretation: Wastage cost is incurred only when forecast exceeds actual demand (the max operator ensures this). Each excess unit is penalized by (a) the expected loss from expiration and (b) the daily cost of keeping it in inventory.

6.3 6.3. Stockout Cost Function (C_s)

When actual demand exceeds the forecast ($A_t > F_t$), unfilled prescriptions generate stockout costs. The stockout cost for drug i on day t is:

$$C_s^{(i)}(t) = \max(A_t^{(i)} - F_t^{(i)},\ 0) \times (\alpha \cdot c_{\text{unit}}^{(i)} + c_{\text{emergency}}^{(i)} + c_{\text{churn}}^{(i)})$$

where:

- α = **asymmetric penalty multiplier**, set to $\alpha = 10$ by default, reflecting the principle that a stockout is approximately $10\times$ more costly than holding one excess unit. This multiplier encodes the clinical imperative: a patient who cannot fill a prescription for a chronic medication (e.g., Metformin for diabetes, Lisinopril for hypertension) faces direct health consequences.
- $c_{\text{emergency}}^{(i)}$ = cost of an emergency reorder (expedited shipping, typically \$15–\$50 per order),
- $c_{\text{churn}}^{(i)}$ = estimated revenue loss from patient switching to a competitor pharmacy (lifetime value impact, amortized per incident; typically \$5–\$25).

Interpretation: Stockout cost is incurred only when actual demand exceeds forecast. The α multiplier ensures that the model's decision boundary is shifted toward over-ordering rather than under-ordering, prioritizing patient safety.

6.4 6.4. Total Asymmetric Loss Function

The total daily cost for drug i combines both components:

$$\mathcal{L}^{(i)}(t) = C_w^{(i)}(t) + C_s^{(i)}(t)$$

And the aggregate cost across all D drugs over a planning horizon of T days is:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^{D} \sum_{t=1}^{T} \mathcal{L}^{(i)}(t)$$

The optimization objective is to find an **order quantity** $Q_t^{(i)}$ for each drug and day that minimizes $\mathcal{L}_{\text{total}}$. Since the stockout penalty ($\alpha = 10$) heavily dominates the wastage penalty, the optimal order quantity will tend toward the **upper prediction interval** of the Prophet forecast rather than the point forecast—a behavior that is clinically appropriate.

6.5 6.5. Practical Example

Consider Amoxicillin 500mg with the following parameters:

Parameter	Value
Unit cost (c_{unit})	\$0.50
Expiration probability (p_{exp})	0.05
Daily holding cost (c_{hold})	\$0.03
Emergency reorder cost ($c_{\text{emergency}}$)	\$25.00
Patient churn cost (c_{churn})	\$10.00
Asymmetric multiplier (α)	10

Scenario A — Over-prediction by 10 units:

$$C_w = 10 \times (0.50 \times 0.05 + 0.03) = 10 \times 0.055 = \$0.55$$

Scenario B — Under-prediction by 10 units:

$$C_s = 10 \times (10 \times 0.50 + 25.00 + 10.00) = 10 \times 40.00 = \$400.00$$

The stockout cost (\$400.00) is **727× larger** than the wastage cost (\$0.55), validating the asymmetric design: the system will strongly prefer slight over-ordering to any risk of under-ordering.

7 7. Technical Methodology

7.1 7.1. System Architecture

The project follows an ETL (Extract, Transform, Load) and Inference pipeline, now extended with a cost-optimization layer that translates Prophet forecasts into cost-minimized order quantities.

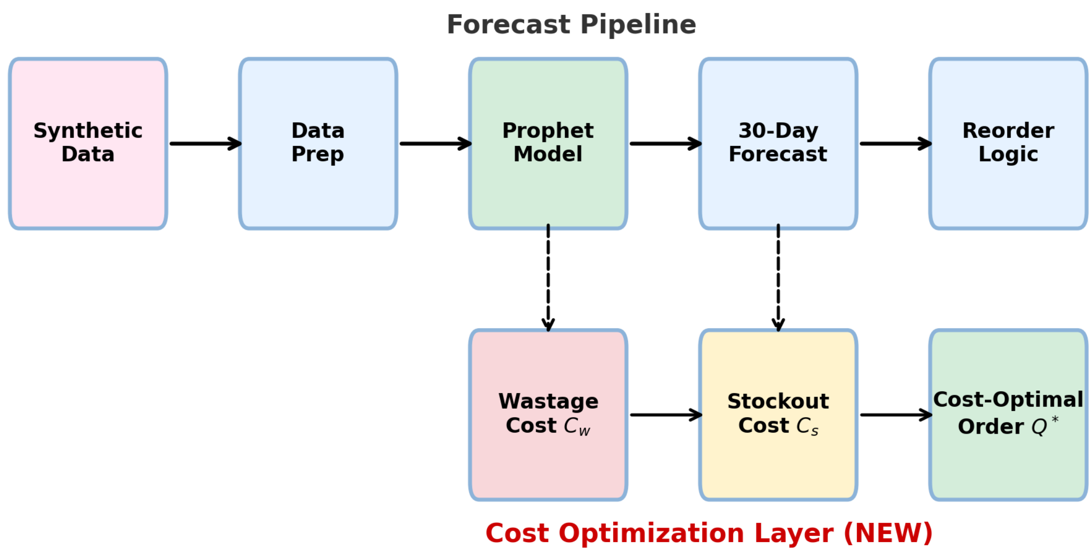

Figure 2: Extended Architecture: Forecast to Cost-Optimized Reorder

7.2 7.2. From Forecast to Cost-Optimal Decision

The integration between Prophet and the cost functions operates as follows:

1. **Prophet generates a probabilistic forecast** for each drug i over a 30-day horizon, producing a point estimate $\hat{y}_t^{(i)}$ and prediction intervals $[\hat{y}_{\text{lower}}^{(i)}, \hat{y}_{\text{upper}}^{(i)}]$ at the 80% confidence level.
2. **The cost functions evaluate candidate order quantities.** Rather than ordering exactly $\hat{y}_t^{(i)}$ (the point forecast), the decision engine evaluates the expected cost $\mathcal{L}^{(i)}$ at multiple candidate quantities between $\hat{y}_{\text{lower}}$ and $\hat{y}_{\text{upper}}$.
3. **The optimal order quantity** $Q_t^{*(i)}$ is the value that minimizes total expected cost:

$$Q_t^{*(i)} = \arg \min_{Q \in [\hat{y}_{\text{lower}}, \hat{y}_{\text{upper}}]} \mathbb{E}[\mathcal{L}^{(i)}(Q, A_t)]$$

Because $\alpha = 10$, this optimization will consistently select quantities **above the median forecast**—typically near the 80th–90th percentile of the predicted demand distribution. This is the mathematical mechanism by which the system prioritizes patient safety over cost minimization.

4. **The reorder engine** aggregates $Q_t^{*(i)}$ across the lead-time window and applies the urgency classification (CRITICAL / HIGH / MEDIUM / LOW) from the existing decision engine.

7.3 7.3. Model Selection: Facebook Prophet

Facebook Prophet is selected for its strengths in this domain:

- **Additive decomposition** of trend, yearly seasonality, weekly seasonality, and holiday effects aligns naturally with pharmacy demand patterns.
- **Robust to missing data** and outliers—important for pharmacies that may have data gaps on holidays or during stockouts.
- **Prediction intervals** are generated natively, providing the distributional information required by the cost functions in Section 6.
- **Interpretable components** allow pharmacists to understand *why* the model recommends a particular order quantity (e.g., "flu season effect + Monday post-weekend surge").

7.4 7.4. Evaluation Framework

Model performance will be assessed on two complementary axes:

Axis	Metric	Target
Forecast Accuracy	MAPE, MAE, RMSE	MAPE $<$ 20% for chronic drugs
Business Cost	$\mathcal{L}_{\text{total}}$ (asymmetric loss)	Minimize total cost over 30-day horizon
Service Level	Fill rate (% of demand met)	$>$ 95%

This dual evaluation ensures that the model is not just accurate in a statistical sense, but also **cost-effective** in a business sense.

8 8. Timeline & Deliverables

Week	Deliverable
1–2	Synthetic data generation; EDA and feature engineering
3–4	Prophet model training; baseline MAPE evaluation
5–6	Cost function implementation; asymmetric loss integration
7–8	Decision engine with cost-optimal reorder logic
9–10	Streamlit dashboard; final evaluation and documentation

9 9. References

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